

Task

Islam's Carpet Cleaning Service

Charges:

\$25 per small

\$35 per large

Sales tax rate is 6%

Estimates are valid for 30 days

Prompt the user for the number of small and large rooms they would like cleaned

and provide an estimate such as:

Estimate for carpet cleaning service

Number of small carpets: 3 <--

Number of large carpets: 1 <--

Price per small carpet : \$25

Price per large carpet : \$35

Cost : \$110

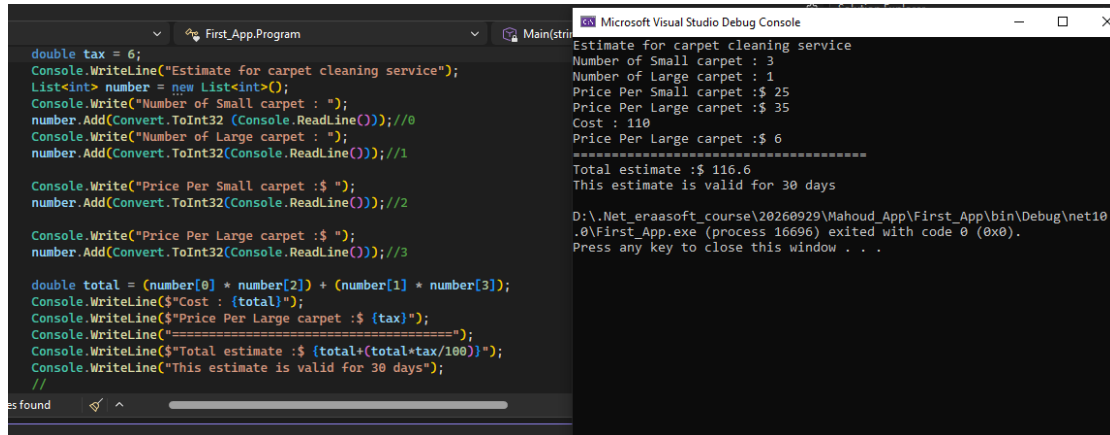
Tax: \$6.6

=====

Total estimate: \$116.6

This estimate is valid for 30 days

The Answer



The screenshot shows a Visual Studio IDE with a C# program in the left pane and its output in the right pane. The code defines a list of carpet counts, calculates a total cost with a 6% tax, and prints the results. The console output shows the program's execution, including the estimated cost of \$116.6 and a 30-day validity period.

```
double tax = 6;
Console.WriteLine("Estimate for carpet cleaning service");
List<int> number = new List<int>();
Console.Write("Number of Small carpet : ");
number.Add(Convert.ToInt32(Console.ReadLine()));//0
Console.Write("Number of Large carpet : ");
number.Add(Convert.ToInt32(Console.ReadLine()));//1

Console.Write("Price Per Small carpet :$ ");
number.Add(Convert.ToInt32(Console.ReadLine()));//2

Console.Write("Price Per Large carpet :$ ");
number.Add(Convert.ToInt32(Console.ReadLine()));//3

double total = (number[0] * number[2]) + (number[1] * number[3]);
Console.WriteLine($"Cost : {total}");
Console.WriteLine($"Price Per Large carpet :$ {tax}");
Console.WriteLine("=====");
Console.WriteLine($"Total estimate :$ {total+(total*tax/100)}");
Console.WriteLine("This estimate is valid for 30 days");
//
```

Microsoft Visual Studio Debug Console

```
Estimate for carpet cleaning service
Number of Small carpet : 3
Number of Large carpet : 1
Price Per Small carpet :$ 25
Price Per Large carpet :$ 35
Cost : 110
Price Per Large carpet :$ 6
=====
Total estimate :$ 116.6
This estimate is valid for 30 days

D:\Net_eraasoft_course\20260929\Mahoud_App\First_App\bin\Debug\net10
.\First_App.exe (process 16696) exited with code 0 (0x0).
Press any key to close this window . . .
```

double tax = 6;

Console.WriteLine("Estimate for carpet cleaning service");

List<int> number = new List<int>();

Console.Write("Number of Small carpet : ");

number.Add(Convert.ToInt32 (Console.ReadLine()));//0

Console.Write("Number of Large carpet : ");

number.Add(Convert.ToInt32(Console.ReadLine()));//1

Console.Write("Price Per Small carpet :\$ ");

number.Add(Convert.ToInt32(Console.ReadLine()));//2

Console.Write("Price Per Large carpet :\$ ");

number.Add(Convert.ToInt32(Console.ReadLine()));//3

double total = (number[0] * number[2]) + (number[1] * number[3]);

Console.WriteLine(\$"Cost : {total}");

Console.WriteLine(\$"Price Per Large carpet :\$ {tax}");

Console.WriteLine("=====");

Console.WriteLine(\$"Total estimate :\$ {total+(total*tax/100)}");

Console.WriteLine("This estimate is valid for 30 days");

Search TASK

1. why the output of this Equation = \$30.00?

لان ال c or :C ترمز الي تحويل الرقم لعملة البلد حيث ظهرت علامة استفهام أي ج م

```
int X = 10;
int Y = 20;
Console.WriteLine($"Equation: {X} + {Y} = {X + Y:C}");
```

Microsoft Visual Studio Debug Console

Equation: 10 + 20 = 30.00 ٣٠.٠٠.?

D:\.Net_eraasoft_course\20260929\Mahoud_App\First_App\bin\Debug\net10.0\First_App.exe (pr
(0x0).

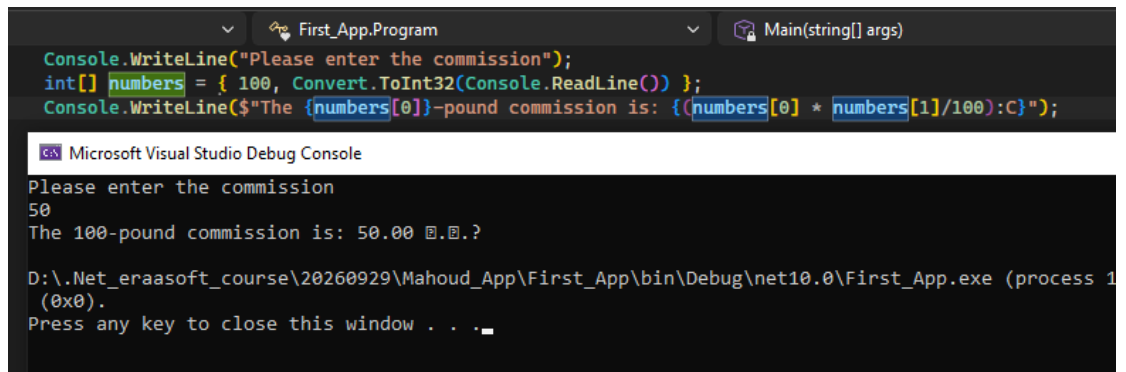
Press any key to close this window . . .

2. what is its benefit?

هناك رموز مختصر يمكن من خلالها تحديد شكل ال format للارقام كما في الجدول التالي ولكن عند محاولة استخدام اكثر من اختصار في نفس السطر يظهر خطأ

الرمز	الاسم	الوصف	مثال الكود (#C)	النتيجة (افتراضياً)
C أو c	Currency	تحويل الرقم لعملة بناءً على الثقافة الحالية (CultureInfo).	(1234.5).ToString("C2")	\$1,234.50 أو 1,234.50 ج.م
D أو d	Decimal	أعداد صحيحة فقط. الرقم يحدد إجمالي الخانات مع إكمال الباقي بأصفار على اليسار.	(42).ToString("D5")	42
E أو e	Exponential	كتابة الرقم بالصيغة العلمية الأسية.	(1052.03).ToString("E2")	1.05E+03
F أو f	Fixed-Point	تثبيت عدد الخانات العشرية بعد الفاصلة مع التقريب تلقائياً.	(1234.567).ToString("F2")	1234.57
G أو g	General	التنسيق العام؛ يختار تلقائياً الشكل الأكثر اختصاراً (ثابت أو علمي).	(1234.5).ToString("G")	1234.5
N أو n	Number	عرض الرقم مع فواصل الآلاف (Group Separator) وخانات عشرية.	(1234567.89).ToString("N2")	1,234,567.89
P أو p	Percent	ضرب الرقم في ١٠٠ وإضافة علامة النسبة المئوية %.	(0.175).ToString("P1")	17.50%
X أو x	Hexadecimal	تحويل الأرقام الصحيحة للنظام الست عشري (Hex).	(255).ToString("X2")	FF

3. try another example with a different specifier with a screenshot of the output.



The screenshot displays the Visual Studio IDE with a C# program being debugged. The code in the editor defines an array 'numbers' with a fixed value of 100 at index 0 and a user-provided value at index 1. It then calculates and prints the commission as a percentage of the 100-pound value. The debug console below shows the program's execution: it prompts for input, receives '50', and outputs 'The 100-pound commission is: 50.00 %'.

```
First_App.Program Main(string[] args)
Console.WriteLine("Please enter the commission");
int[] numbers = { 100, Convert.ToInt32(Console.ReadLine()) };
Console.WriteLine($"The {numbers[0]}-pound commission is: {(numbers[0] * numbers[1]/100):C}");
```

Microsoft Visual Studio Debug Console

```
Please enter the commission
50
The 100-pound commission is: 50.00 %.?

D:\.Net_eraasoft_course\20260929\Mahoud_App\First_App\bin\Debug\net10.0\First_App.exe (process 1
(0x0)).
Press any key to close this window . . .
```